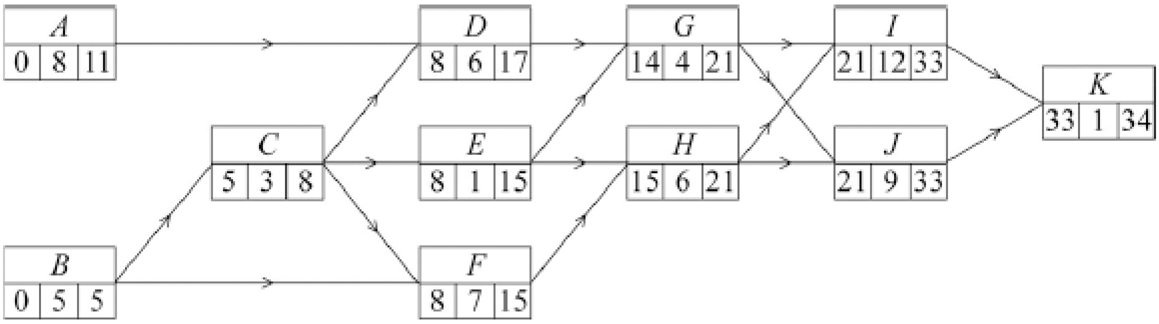
Mark schemes

**Q1.**

(a)




*Forward pass, correct at two of* $D$, $E$, $F$

M1

*All correct*

A1

*Backward pass, correct at* $G$ ***AND*** $H$ *ft*

M1

*All correct*

A1

**4**

(b)

| Activity | Predecessor |
|---|---|
| $A$ | – |
| $B$ | – |
| $C$ | $B$ |
| $D$ | $A$, $C$ |
| $E$ | $C$ |
| $F$ | $B$, $C$ |
| $G$ | $D$, $E$ |
| $H$ | $E$, $F$ |
| $I$ | $G$, $H$ |
| $J$ | $G$, $H$ |
| $K$ | $I$, $J$ |

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*6+ correct*

B1



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*All correct*

B1
**2**

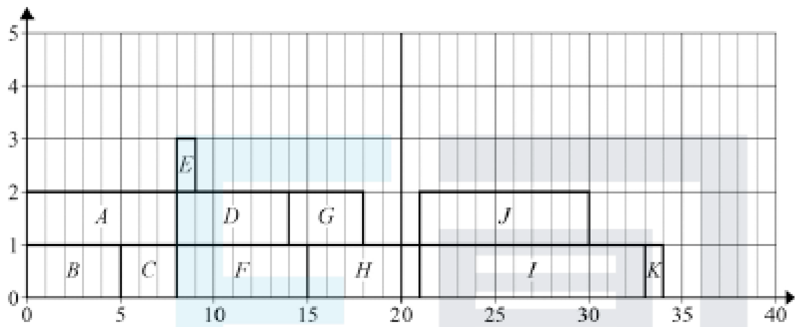
(c) (Critical) $B\ C\ F\ H\ I\ K$

B1
**1**

(d) (Float $E$) 6 (hrs)

B1
**1**

(e)




*Their critical activities and 3 others shown*

M1

*Critical activities and 3 others correct*

A1

*All correct, condone floats seen*

A1
**3**

(f) 34 (hrs)

B1
**1**

(g) 62 (hrs)

B1
**1**

**[13]**

**Q2.**
(a)



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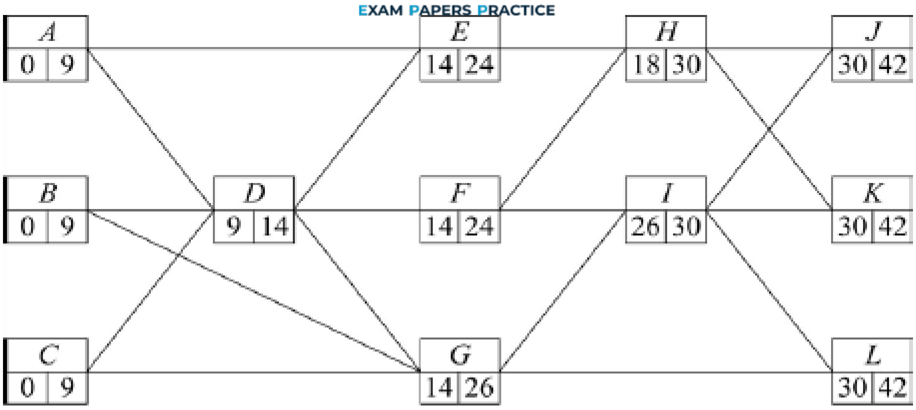


*Forward pass, correct at $D, E, F, G$*

M1

*All correct*

A1

*Backward pass, correct at $H, I, G$ ft*

M1

*All correct*

A1

**4**

(b) $C\ D\ G\ I\ J$ only

B1

**1**

(c) 6

*Their (latest – earliest – 4)*

B1ft

**1**

(d) $H$ delayed by 4

E1

$K$ delayed by 5

B1

New time 51

*51 scores 3/3*

B1

**3**

**[9]**

## Q3.

(a) $x = 4$

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B1

$y = 12$

B1

$z = 13$

B1

**3**

(b) $B\ D\ H\ J$ and $C\ E\ I\ J$

*first correct path*

M1

*2nd correct and no others*

A1

**2**

(c) $G$

B1

Float = 3

B1

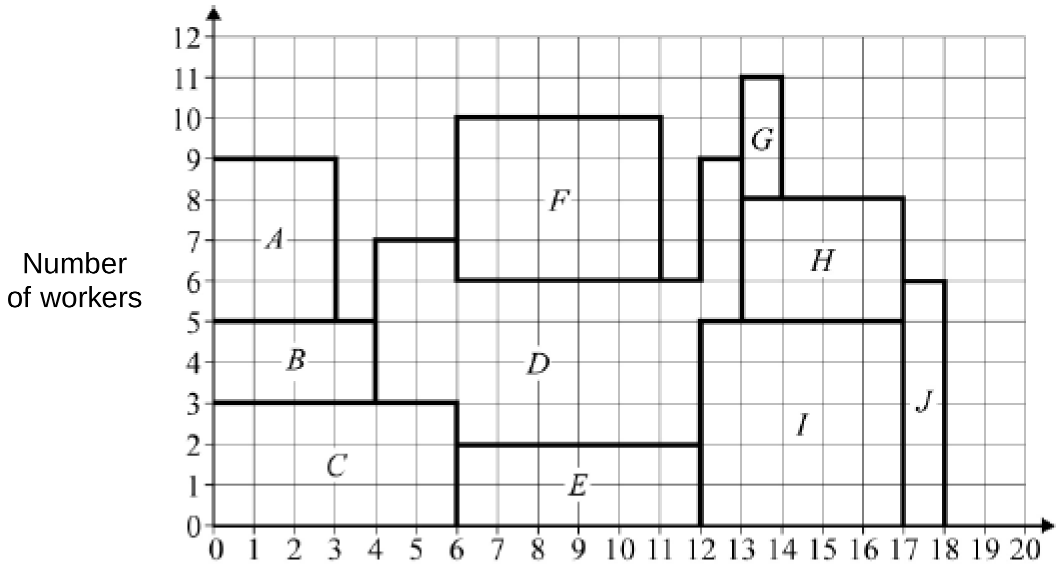
**2**

(d) One of their CPs correct height *and correct durations*

M1

$B$, $D$, $H$, $J$ and $C$, $E$, $I$ correct *and correct durations*

A1

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Number of days

$A$ starting at 0 and ending at 3

*one correct with correct height*

M1

$F$ starting at 6 and ending at 11

*two correct with correct height*

A1

$G$ starting at 13 and ending at 14

*all correct with correct height*
*withhold first A1 earned if it is not clear which activities take place at any given time*
*withhold another A1 if "holes" appear in histogram*

A1

**5**

(e) New earliest $J$ 22 days

*assuming activities continuous*

B1

Minimum extra time 5 days

*assuming activities continuous*

B1

**2**

**[14]**

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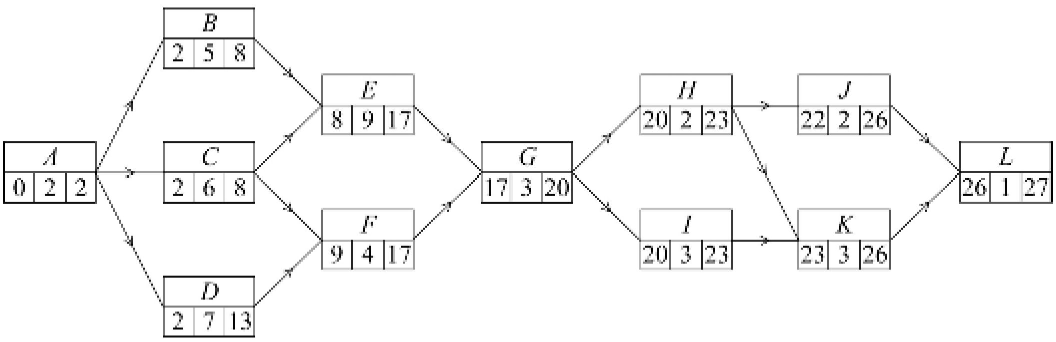
**Q4.**




(a) Forward pass

*up to one slip ft*

M1



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Correct

A1

Backward pass

*up to one slip ft*

M1

Correct

A1

**4**

(b) (i) Critical path $A\ C\ E\ G\ I\ K\ L$

B1

(ii) Float for $D$ = 13 – 2 – 7

*'their 13' − 'their 2' − 7*

M1

= 4 days

A1

**3**

(c)

$A\ C\ E\ G\ IK\ L$ correct durations and heights

*one slip in duration or height*

M1

*correct*

A1

$D$ and $B$ and $F$ correct (no "holes")

B1

$H$ and $J$ correct (no "holes")

*withhold final mark earned if not clear which activities are taking place at any time*

B1

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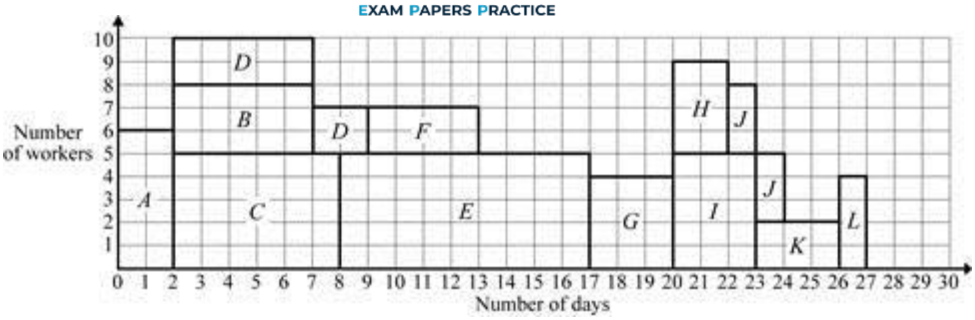



4

(d) Correctly dealing with $D$, $B$ and $F$

*ft 1 slip*

B1

Correctly dealing with $H$ and $J$

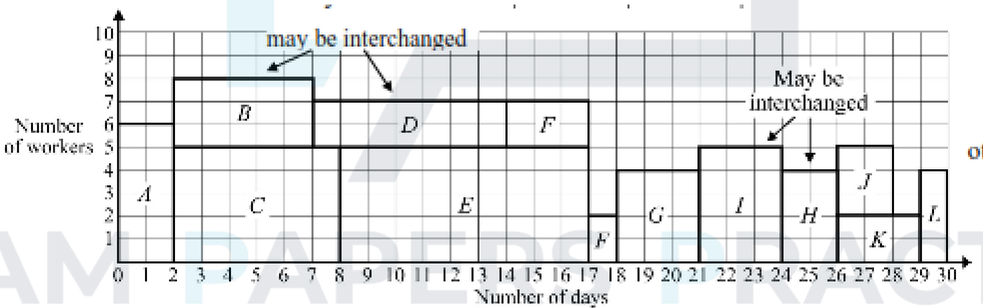
*ft 1 slip*

B1

Minimum extra time = 3 days

*CAO*

B1




other possibilities

3

**[14]**

## Q5.

(a)

| Activity | Immediate Predecessors |
|---|---|
| $A$ | – |
| $B$ | – |
| $C$ | $A$, $B$ |
| $D$ | $B$ |
| $E$ | $B$ |
| $F$ | $C$ |
| $G$ | $D$ |
| $H$ | $D$, $E$ |
| $I$ | $F$, $G$ |

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| | |
|---|---|
| $J$ | $G, H$ |
| $K$ | $I, J$ |

*Up to 2 slips*

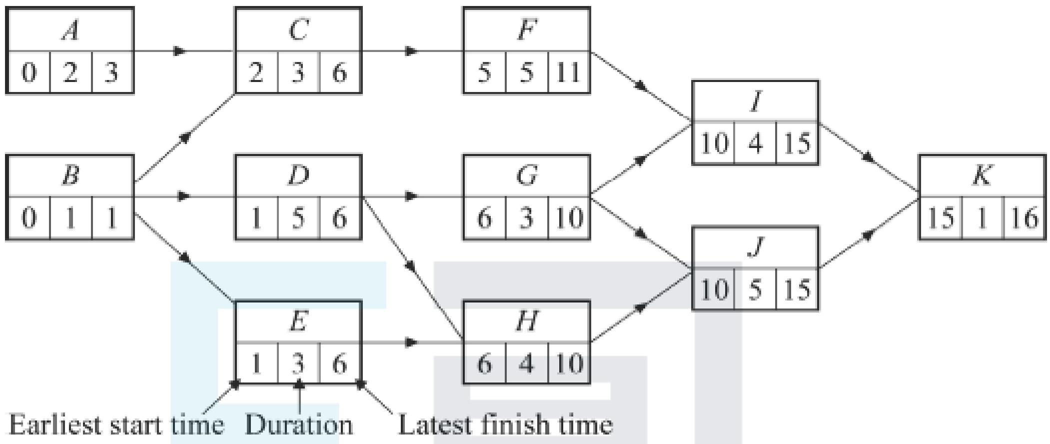
M1

*All correct*

A1

**2**

(b)




*Start times – up to 1 slip with FT*

M1

*All correct*

A1

*Finish times – up to 1 slip; FT 'their 16'*

M1

*All correct; CSO*

A1

**4**

(c) Critical path $B\ D\ H\ J\ K$

B1

Minimum time 16 days

B1

**2**

(d) Greatest float at $E$

B1ft

Value = 2

B1ft

**2**

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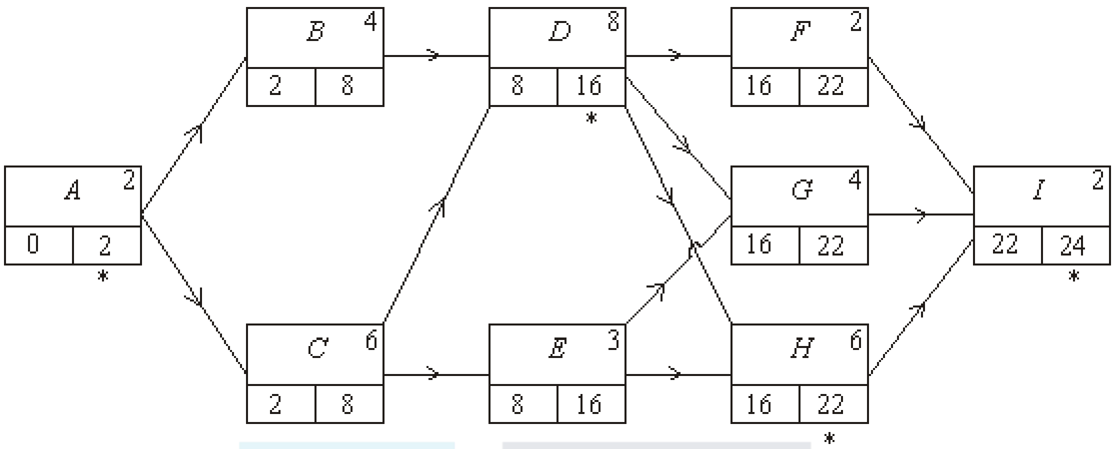
**[10]**

## Q6.

(a)




Activity network SCA

M1

*almost correct (up to 2 slips)*

A1

*all correct*

A1

**3**

(b) Forward pass for earliest times

M1

A1

**2**

(c) Backward pass

M1

A1

**2**

(d) Critical path is ACDHI

B1

Minimum completion 24 days

B1

**2**

(e)

| | | | | |
|---|---|---|---|---|
| Non–critical | $B$ | $E$ | $F$ | $G$ |
| Float | 2 | 5 | 4 | 2 |

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*At least 3 activities and float in one activity ft correct*

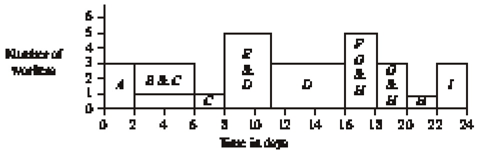
M1

*ft their earliest and latest times*

A1ft

2

(f)




*Histogram ≤ 11*

M1

*Correct*

A1

*Rest as histogram – generally start activities ok*

M1

*All correct*

A1

4

Resource histogram

(g) Problems with $D$ & $E$ solved by $E$ coming after $D$

M1

Problem at 16-18 days with $F$ can be solved by moving $F$ to 20-22

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A1

Must overrun by equivalent to duration of $E$ (3 days)

B1

3

**[18]**

## Q7.

(a)



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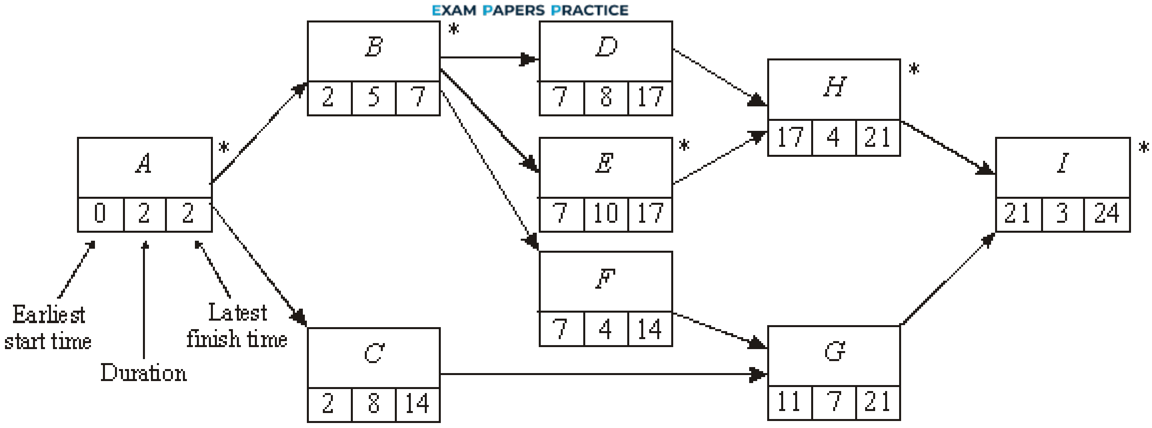


*SCA*

M1

*(almost correct 2 slips)*

A1

*Correct*

A1

**3**

(b) Forward pass for earliest start times

M1

*All correct*

A1

**2**

(c) Backward pass for latest finish times

M1

*All correct*

A1

**2**

(d) Critical path $A\ B\ E\ H\ I$

B1

**1**

(e)

| | | | | |
|---|---|---|---|---|
| Non critical | $C$ | $D$ | $F$ | $G$ |
| Float | 4 | 2 | 3 | 3 |

*At least one float time correct*

M1

*All correct*

A1

**2**

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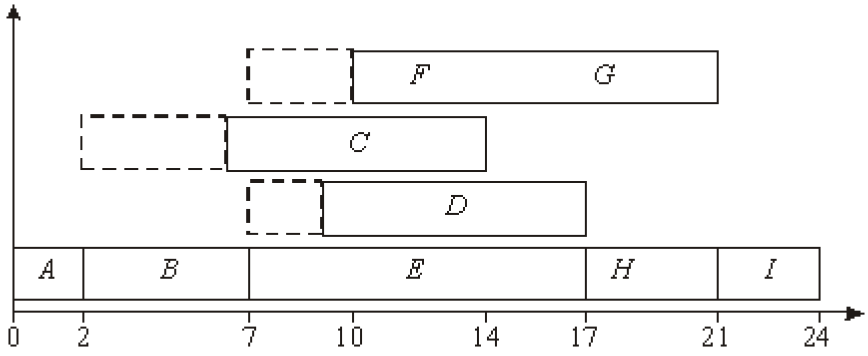
(f)




'their' critical path on chart

B1ft

C from 6 to 14 (with space 2-6)

*One other activity (condone no slack or earliest start)*

M1

D from 9 to 17 (with slack 7-9)

*2 other non critical activities*

A1

F & G from 10 to 21 with appropriate slack

*All correct*

A1

4

**[14]**

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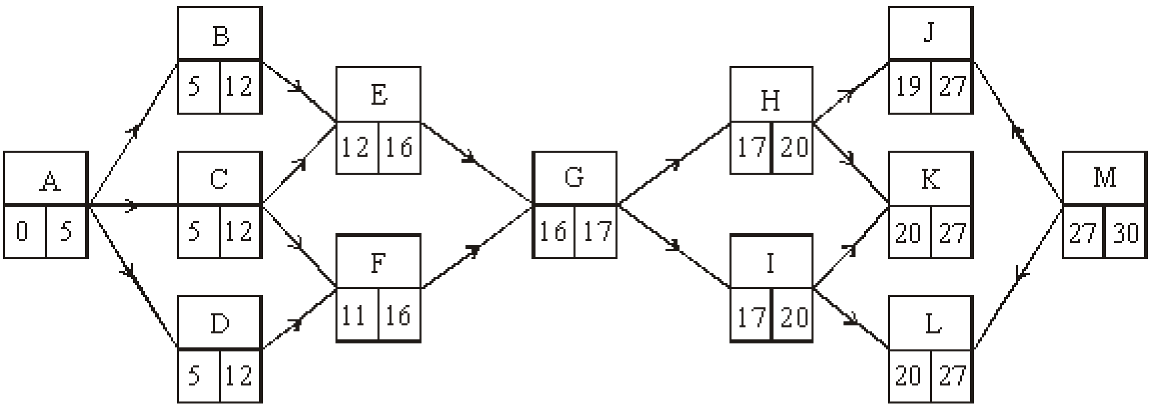
**Q8.**

(a)




*Forward pass*

M1 A1

2

*Back pass*



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M1 A1 **2**

(b) Critical path *ABEGILM*

B1 **1**

(c) *D*

B1 **1**

(d) (*C* now 9, so *E* starts at 14)

*SC 32 scores ½*

M1

Overrun 2 days

A1 **2**

(e) *F* start at 14

*SC 35 scores 2/3*

M1A1

∴ overrun 5 days

A1 **3**

**[11]**

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**Q9.**

(a)

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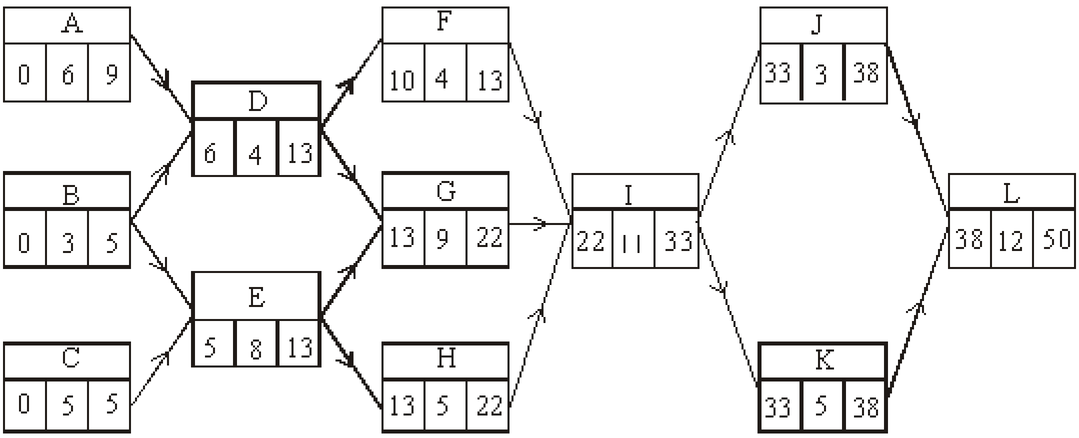


*forward pass*

M1A1 **2**

*back pass*



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M1A1

2

(b) $C\ E\ G\ I\ K\ L$

B1

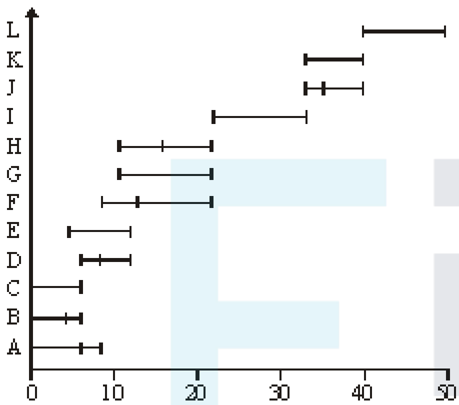
1

(c) $F$

B1

1

(d)




*Gantt diagram*

M1

*floats included*

B1

*correct, excluding floats*

A1

3

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(e) (i) $D$ 5 days $\Rightarrow$ $G$ back 2 days

E1

$\therefore$ $J$ starts at 35

*OE*

M1

$\therefore$ $L$ starts at 43

$\therefore$ finish at 55

A1

3

(ii) $A\ D\ G\ I\ J\ L$

B1



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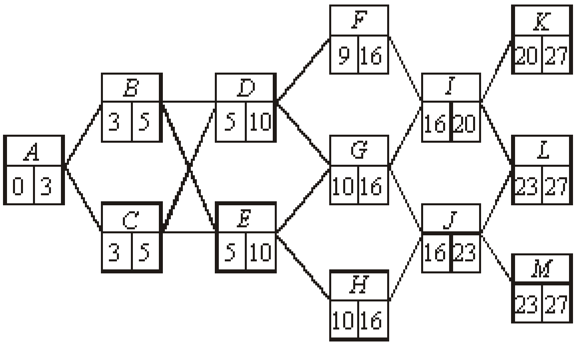
1

**[13]**

## Q10.

(a) (i) & (ii)




*Forward*

M1

A1

2

*Back*

M1

A1

2

(b) $C, H, F, D$

B1

1

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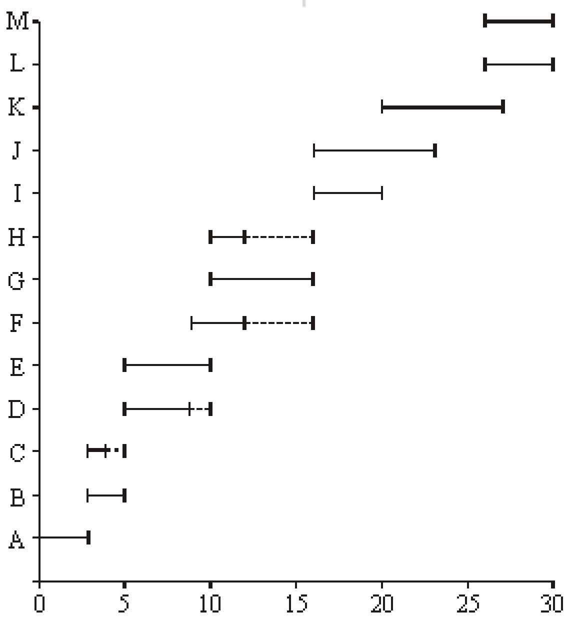
(c)



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M1

*(–1 EE)*

A2

**3**

(d) (i) Extra at $KLM$

M1

Min extra 4
Total = 31

A1

**2**

(ii) $A$ $B$ $D$ $F$ $H$ $I$ $K$
$C$ $E$ $G$ $J$ $L$ $M$

B1

*OE*

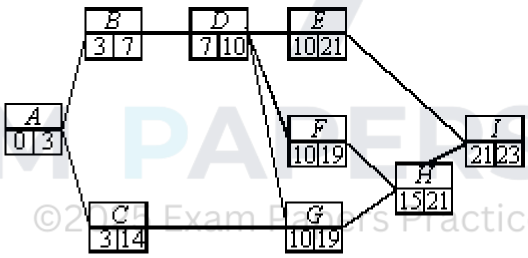
B1

**2**

**[12]**

## Q11.

(a/b/c)




*SCA*

M1

*–1 EE*

A2

**3**

*for forward pass*

M1

*for all correct*

A1

**2**

*for back pass*

M1

*for all correct*

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A1

2

(d) $C - 9$
$F - 5$
$G - 4$
$H - 4$

2

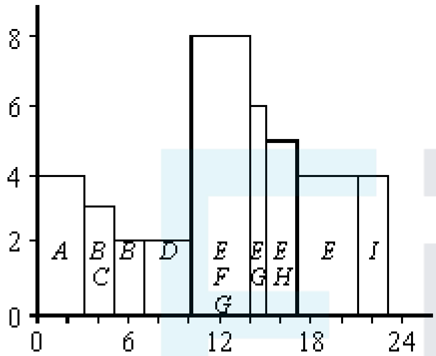
*for all fours letters*

B1F

*for four correct floats*

B1

(e)




M1

*SCA*

M1

*–1EE*

A3F

4

**[13]**

**Q12.**

(a)

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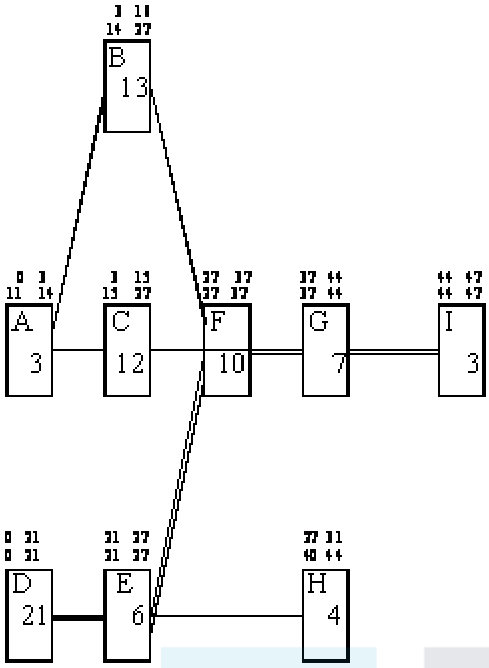


*Attempt at a network with all 9 activities included, starting at A and D, finishing at I*

M1

*All correct, – 1 EE*

A3

**4**

(b) Upper figures show earliest times
*Evidenced by labels*

M1

(forward scan)
*All correct*

A1

Lower figures show latest times
*Evidenced by labels*

M1

(backward scan)
*All correct*

A1

Critical path *D-E-F-G-I*
*Critical path stated and shown. ft c's labels provided upper and lower figures match*

A1FA1F

**6**

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(c)

| *A* | *B* | *C* | *D* | *E* | *F* | *G* | *H* | *I* |
|---|---|---|---|---|---|---|---|---|
| 11 | 11 | 12 | 0 | 0 | 0 | 0 | 13 | 0 |

*– 1 EE*

B2 2

(d) Cascade diagram

*Attempt a cascade diagram with all activities and a suitable time scale*

M1

*All correct. – 1 EE*

A3

Minimum number of helpers: 3

*No. of helpers (can be earned independently).*

B1 5

(e) Least delay: 1 day

*Delay*

B1

Achieved by starting *A* and *B* as early as possible
then do *C* immediately after *B*

*OE involving B and C sequential rather than parallel*

E1 2

**[19]**

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Examiner reports

## Q1.

This question proved to be a good source of marks for all students, with many scoring full marks. There were problems in part (e) where some students left ‘holes’ with their blocks and others had the correct histogram profile, but failed to indicate the activities linked to each block.

## Q2.

The first three parts of this question proved to be a good source of marks for all students with many scoring full marks. There were problems in part (d), where students failed to appreciate the effect of BOTH $H$ and $K$ overrunning and the implications to the critical path.

## Q3.

Part (a) Almost every candidate found the correct values of the constants $x$, $y$ and $z$.

Part (b) The two critical paths were found correctly by most candidates, but several candidates wrote down extra paths that were not critical paths.

Part (c) Most candidates identified the correct activity with the largest float and calculated the float correctly.

Part (d) The histogram was usually constructed correctly but a few candidates left “holes” in their blocks and others made slips having certain blocks with an incorrect height or width.
It was necessary to indicate clearly which activities were taking place at any given time; those who had the correct histogram profile, but without the activities indicated, did not score full marks.

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Part (e) The resource levelling caused difficulties for many, with only the best candidates finding the correct minimum extra time required and the new start time for activity $J$.

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## Q4.

In part (a) almost every candidate calculated the earliest start times and latest finish times correctly.

In part (b)(i) the single critical path was again found correctly by almost everyone.

In (b)(ii) a few candidates did not know how to calculate the float for activity $D$, but the majority were able to do so by finding 13 – 2 – 7.

In part (c) the histogram was usually constructed correctly but a few candidates left gaps or “cantilevers” in their blocks and others made slips having certain blocks with an incorrect height or width. It was pleasing to see most candidates indicating clearly which activities were taking place at any given time.

In (d) the resource levelling caused a few problems; some could not deal correctly with the activity $F$ which needed to be fitted in before $G$ started. A number of alternatives were possible with the events following $G$, but the common mistake was trying to start $K$ before



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$H$ had finished. Those with a correct second histogram had no trouble in stating the correct minimum extra time required.

## Q5.

The precedence table was usually correct and the earliest start times and latest finish times were found correctly by most candidates. Quite a few gave the latest finish time for activity $F$ as 10 rather than 11, and consequently had values of 5 and 2 at $C$ and $A$ respectively; this was treated as a single error. Those with the correct values on the activity diagram were able to find the correct critical path and were able to conclude that activity $E$ had the greatest float time. This proved to be a very good opening question for all candidates.

## Q6.

Parts (a),(b),(c),(d) and (e) were very well done by practically all the candidates. The topic of critical path analysis seems to be well understood by most candidates. Problems started to creep in when answering part (f) where sometimes the resource histogram was not a histogram at all; being more like a random collection of rectangles piled on top of each other, rather like Stonehenge. Candidates had been expected to indicate which activities were being represented on the histogram so as to make resource levelling easier to identify. Many candidates were successful in finding the minimum extra time, although the explanations as to why three extra days are sufficient were often incomplete.

## Q7.

The network diagram and the critical path were answered very well by all candidates. A few were unable to identify float times for the non-critical activities. The major problem in the Gantt chart was a failure to show slack time. When this was indicated it was usually shown after the various activities even though the question asked to show activities starting as **late** as possible. Some indication should have been made on the diagram to indicate that G needed to start after F and also after C but this was not penalised this time.

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## Q8.

This question proved to be a good source of marks for all candidates. A significant number of candidates failed to answer parts (d) and (e) correctly. Correct solutions were sometimes obtained by candidates amending their figure by including the overruns and their implications.

## Q9.

This question proved to be a good source of marks for all candidates, although a significant number failed to score full marks in part (d) due to the inaccuracies of their diagram or through drawing a resource histogram. Part (e) discriminated well between candidates. This was the first time that candidates were provided with a stage/state table for answering a dynamic programming question. The question gave candidates the opportunity to use a network diagram and still score full marks. A significant number of weaker candidates thought that, as there were only six possible orders, a complete enumeration was an acceptable method. This was not the case. About half of the candidates worked backwards through the network and about half worked forwards. Both methods were equally successful.

## Q10.



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This question proved to be a good source of marks for all candidates, although a significant number of candidates failed to score full marks in part (c), due to the inaccuracies of their diagram or through drawing a resource histogram instead of a Gantt diagram.

## Q11.

This question was well answered by the majority of the candidates, with the exception of part (e). The vast majority of candidates drew an 'activity on node' diagram and this proved to be very successful. Part (e) required candidates to draw a resource histogram for the project but unfortunately a significant number of candidates merely drew a Gantt diagram. Candidates should be encouraged on all such diagrams to use graph paper and to make sure their work is well presented.

## Q12.

The question was a routine application of critical path analysis for which the candidate was well prepared and scored nearly all the available marks.

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